

Step selection analysis in three dimensions

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2026-03-04

In this document, we illustrate the workflow of the analysis of (simulated) three-dimensional tracking data using a step selection function. The steps of the analysis are very similar to the two-dimensional case, but we use functions from the package `ssf3D`.

```
library(ssf3D) # for data processing and analysis
library(terra) # for raster stack
library(lubridate) # to process dates
library(survival) # to fit the SSF
```

Data

We start from a three-dimensional movement track, which contains x/y/z coordinates at regular time intervals. Here, we do not cover possible pre-processing steps, for example to obtain temporally regular positions using interpolation or a state-space model. The data frame also has columns for ID (track identifier) and `time`.

```
# Load tracking data
tracks <- read.csv("sim_data.csv")
tracks$time <- ymd_hms(tracks$time)
head(tracks)
```

	ID	x	y	z	time
1	1	270.9550	216.6026	164.5475	2024-01-01 12:00:00
2	1	269.1909	215.2127	161.9610	2024-01-01 13:00:00
3	1	267.9176	213.6112	159.9985	2024-01-01 14:00:00
4	1	264.5435	213.6372	158.1060	2024-01-01 15:00:00
5	1	260.3760	208.8796	154.7549	2024-01-01 16:00:00
6	1	260.7953	207.1210	154.2573	2024-01-01 17:00:00

We also have a three-dimensional environmental variable. This is simulated data, but in a real analysis, this could be something like wind, temperature, or any other spatial variable measured over a three-dimensional grid. In two dimensions, spatial data is typically stored as in raster format. Here, we use a raster stack from the terra package, where the layers range across the third (vertical) dimension. We load the raster stack from a tif file, and set the `depth` attribute to the vertical coordinates of the raster layer (used later to extract).

```
# Load spatial covariate data
r_stack <- rast("3D_raster.tif")
depth(r_stack) <- 1:nlyr(r_stack)
r_stack
```

```
class      : SpatRaster
size       : 500, 500, 300  (nrow, ncol, nlyr)
resolution : 1, 1  (x, y)
extent     : 0, 500, 0, 500  (xmin, xmax, ymin, ymax)
coord. ref.:
source     : 3D_raster.tif
names      :          1,          2,          3,          4,          5,          6, ...
min values : 0.1060584, 0.1269960, 0.1301358, 0.1235008, 0.1015807, 0.0444927, ...
max values : 0.8853686, 0.8797945, 0.9235040, 0.9013632, 0.9261275, 0.9409347, ...
depth      : 1 to 300 (300 steps)
```

Generate random points

We use the function `pred_data()` from `ssf3D` to generate random points for the step selection analysis. This is analogous to the random (or “control”) points used in two dimensions, but here they are three-dimensional points. We specify `distr = "gamma"` to indicate that distances between the start point of each step and the random points follow a gamma distribution (for which parameters are estimated from the observed data). By default, random points are generated with uniform bearings.

```
# Generate random points and get movement variables
data <- prep_data(data = tracks, n_random = 50, distr = "gamma")
data[1:5, 1:6]
```

	ID	stratum	obs		x	y	z
1	1	1-3	1	267.9176	213.6112	159.9985	
2	1	1-3	0	270.9445	216.3967	163.9238	
3	1	1-3	0	269.5533	214.7839	163.0450	

```
4 1      1-3    0 273.2034 213.9234 168.1376
5 1      1-3    0 271.3224 216.5781 161.4422
```

The new data frame has columns for `stratum`, which groups each observed position with matched random points, and `obs`, which is 1 for observed positions and 0 for random points.

In addition to generating random points, the function `prep_data()` also calculated all movement variables required for the analysis. In particular, `step` is the step length, `omega` is the geodesic length, and `delta` is the geodesic orientation. We will use those three variables later to model step lengths with a gamma distribution and the geodesics with a Kent distribution.

Get covariates

We need to extract the spatial covariate at all observed and random positions, and we do this using the function `extract_3d()` from the package `ssf3D`. This is similar to the function `extract()` in the packages `raster` and `terra`, but it also uses the vertical coordinate for a covariate measured in three dimensions.

```
# Extract spatial covariate values
data$cov <- extract_3d(r_stack, data[,c("x", "y", "z")])
```

We also calculate the movement variable $\sin(\omega)^2 \cos(2\delta)$, which we will include to model geodesics with a Kent distribution.

```
# Compute transformed variable for Kent distribution
data$omega_delta <- sin(data$omega) * sin(data$omega) * cos(2 * data$delta)
```

Fit step selection function

We fit the step selection function using conditional logistic regression, implemented in the `clogit()` function of the `survival` package. The formula includes `step` and `log(step)` to model step lengths with a gamma distribution, `cos(omega)` and `omega_delta` to model geodesics with a Kent distribution, and `cov` to estimate habitat selection. We also include the term `strata(stratum)` to specify the grouping of observed and random points.

```
fit <- clogit(obs ~ step + log(step) +
              cos(omega) + omega_delta +
              cov +
              strata(stratum),
              data = data)
```

```
fit
```

Call:

```
clogit(obs ~ step + log(step) + cos(omega) + omega_delta + cov +  
      strata(stratum), data = data)
```

	coef	exp(coef)	se(coef)	z	p
step	-2.060e-02	9.796e-01	2.590e-02	-0.795	0.426
log(step)	1.054e-01	1.111e+00	9.273e-02	1.137	0.256
cos(omega)	1.051e+01	3.674e+04	4.257e-01	24.691	< 2e-16
omega_delta	5.534e+00	2.532e+02	2.860e-01	19.352	< 2e-16
cov	2.968e+00	1.946e+01	4.359e-01	6.810	9.78e-12

Likelihood ratio test=3201 on 5 df, p=< 2.2e-16

n= 45900, number of events= 900

The fitted model object includes estimated coefficients and standard errors.

Model outputs

Habitat selection

Movement parameters

Like in two dimensions, the interpretation of movement parameters is not straightforward when random points are not spatially uniform. Indeed, the estimated parameters measure a deviation between the distribution of the random points and the distribution of observed movement variables. The `ssf3D` package includes a function to correct the estimates to account for the distribution of random points, and a function to obtain the estimated parameters of the implied gamma distribution.

```
beta <- correct_betas(model = fit, data = data)  
beta
```

step	log(step)	cos(omega)	omega_delta	cov
-0.4972962	-0.9626594	10.5117226	5.5343471	2.9683054

```
gamma_par <- transform_betas(beta_step = beta["step"], beta_logstep = beta["log(step)"])  
gamma_par
```

shape.log(step)	scale.step	mean.log(step)	sd.log(step)
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2.037341

2.010874

4.096835

2.870230